QP Code. 25900911

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION, NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCPHY100 – FOUNDATIONS OF PHYSICS

Duration: 1 hour 30 minutes **Maximum Marks:50**

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

PART A

Short Answer Questions
Answer any 7 Questions
Each Question carries 2 marks

1. Find the magnitude of the vector $\boldsymbol{A}= 6i - 3j + 2k$. [A]/[CO1]
   Ans. $Magnitude\ of\ A = \sqrt{6^2 + (-3)^2 + 2^2} = 7$

   2 marks for correct answer.

2. Give examples of significant figures. [K]/[CO1]
   Ans. *Any two examples based on the rule for significant figures.*

   2 marks for 2 examples. 1 mark for 1 example.

3. Which quantity is obtained from the area of velocity-time graph? Explain. [U]/[CO3]
   Ans. *Area under velocity-time graph gives displacement*

$$\int_{t_1}^{t_2} vdt = \int_{x_1}^{x_2} dx = x_2 - x_1$$

   Give one mark for writing displacement alone
   Either graphical explanation or representation of equation-give one mark

4. What is relative velocity? [K]/[CO3]
   Ans. In general, when two observers measure the velocity of the same body, they get different results if one observer is moving relative to the other. The velocity seen by a particular observer is called the velocity *relative* to that observer, or simply **relative velocity.**
   2 marks for explanation. 1 mark for equation $\boldsymbol{v_{b/a} = v_b - v_a}$ alone

5. Explain why mass is considered a scalar quantity, while weight is a vector quantity. [U]/[CO3]

   Ans. *Mass is the amount of substance or matter available in any object whereas weight is the force experienced by a mass $'m'$ due to gravity. Force being a vector weight is also a vector.*

   $$\boldsymbol{F = mg}$$

   2 marks for explanation. 1 mark for equation alone.

6. Define conservative force. Give an example for conservative force. [U]/[CO4]
   Ans. A conservative force is one for which the work–kinetic energy relationship is completely reversible. The work done by a conservative force around any closed path is zero. or
   A conservative force is a force whose work depends only on the initial and final positions of an object, not on the path taken.
   Examples: Gravitational force, spring force etc.

   1 mark for any explanation. 1 mark for example.

7. What are stable equilibrium and unstable equilibrium positions? [K]/[CO4]
   Ans. A body is in *stable equilibrium* if, when slightly displaced, it experiences a restoring force or torque that brings it back to its original position. In potential energy curve it is the minimum energy position.
   Example: A ball resting at the bottom of a bowl. If nudged, it rolls back to the bottom.

   A body is in *unstable equilibrium* if, when slightly displaced, it experiences a force or torque that moves it further away from its original position. In potential energy curve it is the maximum energy position.
   Example: A ball balanced at the top of a hill. If nudged, it rolls down and never returns to the top.
   1 mark for stable equilibrium and 1 mark for unstable equilibrium.
   0.5 mark each for examples alone.
   Representation via graph- give 1 mark

8. Describe the weight of a body? Is it a constant? [U]/[CO3]

   Ans. *Weight is the force experienced by a mass $'m'$ due to gravity.* $\boldsymbol{F = mg}$

As $'g'$ varies, Weight will also vary. For example, on the moon $'g'$ is very small, so weight of an object of mass $'m'$ will be smaller compared to its weight on Earth.

Description of weight 1 mark. 0.5 mark for equation alone. 1 mark for explaining whether weight is a constant.

9. What is the difference between input and raw input? [U]/[CO5]
   Ans. **Give full marks for any reasonable attempt on what input is or anything related to raw input**

10. What is a comment in python programming? [K]/[CO5]
    Ans. Comment is a line of text in the code begins with the # symbol that is used to explain what the code does, make notes for future reference, or temporarily disable parts of the code during testing.
    Correct Description- 2 marks.

Part B
Short Essay Questions
Answer any 4 Questions
Each Question carries 6 marks

11. Define projectile motion. What are the assumptions made in the study of projectile motion? [K]/[CO3]

    Ans. A **projectile** is a body that is given an initial velocity and then follows a path determined entirely by the effects of gravitational acceleration and air resistance. To analyse the motion of a projectile, we represent the projectile as a particle with an acceleration (due to gravity) that is constant in both magnitude and direction. We'll ignore the effects of air resistance and the curvature and rotation of the earth. Projectile motion is always confined to a vertical plane determined by the direction of the initial velocity- i.e. projectile motion is *two-dimensional. Projectile motion can be analysed as a combination of horizontal motion with constant velocity and vertical motion with constant acceleration.*
    2 marks for projectile motion and 4 for assumptions. Give Marks for minimum two assumptions.

12. A particle is observed initially at position (2, 3.5). Three seconds later it is observed at (-3, 5.5) where the distances are expressed in meter. What is the particle's average velocity during this time? [A]/[CO3]

    Ans. $v_{av,x} = \frac{x_2 - x_1}{t_2 - t_1} = \frac{-3-2}{3} = \frac{-5}{3}$; $v_{av,y} = \frac{y_2 - y_1}{t_2 - t_1} = \frac{5.5-3.5}{3} = \frac{2}{3}$

    Average velocity $\boldsymbol{v_{av}} = \left(\frac{-5}{3}, \frac{2}{3}\right)$

Magnitude of average velocity=$\sqrt{\left(\frac{-5}{3}\right)^2+\left(\frac{2}{3}\right)^2}=\frac{\sqrt{29}}{3}$ m/s

2 marks for equation. 3 marks for substitution and calculation. 1 mark for answer with unit. Reduce 0.5 marks if no unit is written.
Alternate methods for calculating average velocity may also be considered

13. Derive the expression for terminal speed in a fluid medium. [U]/[CO3]
Ans. Because of the effects of fluid resistance, an object falling in a fluid does not have a constant acceleration. Let m be the mass of a small ball moving through a fluid. Using Newton's second law,

$$\Sigma F = mg + \left(-kv_y\right) = ma_y$$

When the ball first starts to move, v_y = 0, the resisting force is zero and the initial acceleration is a_y = g. As the speed increases, the resisting force also increases, until finally it is equal in magnitude to the weight. At this time

$$mg - kv_y = 0,$$

the acceleration is zero, and there is no further increase in speed.

The final speed v_t , called the **terminal speed,** is given by

$$mg - kv_t = 0, \text{ or } v_t = \frac{mg}{k}$$

6 marks for deriving this expression. Mentioning of terminal speed alone as a definition give 2 marks

14. What is the difference between positive work, negative work and zero work? [U] / [CO4]
Ans. $W = \boldsymbol{F}.\boldsymbol{s}$ where $\boldsymbol{F}$ and $\boldsymbol{s}$ are force and displacement respectively.

$$W = \boldsymbol{F}.\boldsymbol{s} = Fs\ \cos\theta$$

$$\text{When } \theta = 0^{\circ} \text{and} < 90^{\circ};\ \cos\theta \textit{ is positive. Positive work}$$

$$\text{When } \theta > 90^{\circ} \textit{and less than or} = 180^{\circ};\ \cos\theta \textit{ is negative. Negative work}$$

$$\text{When } \theta = 90^{\circ};\ \cos\theta = 0, W = 0. \textit{Zero work}$$

2 marks each for positive, negative and zero work. 1 mark for equation alone.

15. What are the different forms of energy? [U] / [CO4]
Ans. Defining any two types of energy with relevant equations and explanation- **3 marks each-Total 6 Marks**
Mere mentioning of any two energies with mild explanation give 2 marks.
Just naming of energies – give 1 mark

16. Write a python programme to sum all the elements in the list [10,4,5,7,8,9,11,12]. [A]/[CO5]
Ans.
Define the list

```
numbers = [10, 4, 5, 7, 8, 9, 11, 12]
#  Using built-in sum () function
total = sum(numbers)
print ("Sum of all elements:", total)
Output- Sum of all elements: 66
```

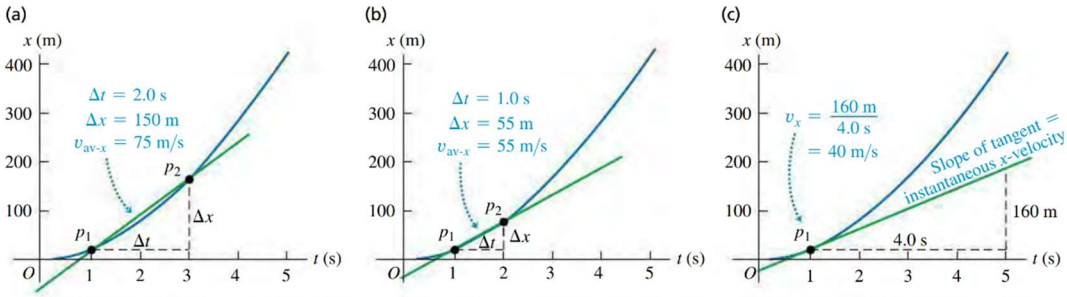
This or any reasonable code which will lead to error free outputs- marks can be given. But assure logic and syntax. 6 marks

Part C
Essay Questions
Answer any 1 Question
Each Question carries 12 marks

17. Discuss in detail, the position-time graph. Describe how average velocity can be calculated. Draw the x-t graph for a body moving with constant velocity. [A]/[CO2]

Ans. Detailed explanation of what position- time graph is, along with graph. **6 marks.** (Just position-time graph alone- **3 marks)**

2.7 Using an *x*-*t* graph to go from (a), (b) average *x*-velocity to (c) instantaneous *x*-velocity v_x. In (c) we find the slope of the tangent to the *x*-*t* curve by dividing any vertical interval (with distance units) along the tangent by the corresponding horizontal interval (with time units).



As the average *x*-velocity $v_{av\text{-}x}$ is calculated over shorter and shorter time intervals ...

... its value $v_{av\text{-}x} = \Delta x/\Delta t$ approaches the instantaneous *x*-velocity.

The instantaneous *x*-velocity v_x at any given point equals the slope of the tangent to the *x*-*t* curve at that point.

Formula for average velocity, $v_{av,x} = \frac{x_2 - x_1}{t_2 - t_1}$. **4 marks for equation with term explanations**

Drawing x-t graph for motion under constant velocity A graph with a straight line . **2 marks.**

18. Explain elastic potential energy and arrive at an expression for it. Also discuss the transformation of energies when a diver jumps off a high board into a swimming pool, assume that air resistance is present. [A]/[CO4]

Ans. Explaining Elastic potential energy with examples or by definition- **4 marks**

Force, $F = kx$

Work done is stored as elastic potential energy; where work done is

$$W = U = \int_0^x F dx = \int_0^x kx dx = \frac{1}{2}kx^2$$

5 marks for obtaining expression of potential energy

On the high board, the energy of the diver is fully gravitational potential energy. When he jumps, gravitational potential energy gets converted to kinetic energy and some heat and sound energy due to the air drag. By the time the diver touches the water, the energy is almost kinetic energy which then is transformed to sound, heat and movement of water.

3 marks